

Client Brief & Internal Context: TechVista Solutions

Company Background:

TechVista Solutions is a rapidly growing enterprise manufacturing company specializing in industrial IoT devices. Over the last three years, they have expanded across 14 different countries in Europe and North America.

Current Architecture & Pain Points:

Currently, their manufacturing plants generate over 500TB of telemetry data daily. However, their existing on-premise data centers are struggling to keep up with the processing load resulting in downtime, delayed alerts, and high maintenance costs.

The current monolithic architecture prevents them from deploying real-time predictive maintenance features. Machine failures often go unnoticed until standard scheduled checks, costing the business roughly \$2.4M annually in unplanned downtime.

Project Goals:

TechVista is seeking a complete digital transformation. They want to migrate to a highly scalable, cloud-native analytics platform (preferably on AWS). They need microservices architecture to process data streams in real-time, generate predictive maintenance alerts, and provide executives with real-time dashboards via QuickSight.

Key Requirements:

- High Availability (HA) across multiple geographical regions.
- Complete data security compliance including end-to-end encryption and IAM role-based access control.
- Integration with Kinesis Data Streams for data ingestion.

Budget Expectations:

The executive board has earmarked approximately \$3M for the entire migration and build phase, with an expected timeline of 6-8 months for minimum viable product rollout.